The Riemann Kernel Is Globally PF_4

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Abstract

Let Φ be the classical Riemann, or de Bruijn–Newman, kernel. We prove that every translation minor of Φ of order at most four is strictly positive at strictly ordered nodes, and an exact origin-confluent calculation gives a negative order-five minor. Thus the exact global Pólya-frequency order is four. The order-five confluent coefficient changes sign exactly once on the positive half-line, at $0.0622795266356 < u_* < 0.0622795266357$, and a rational finite-spacing witness has determinant between $-1.08 \cdot 10^{-7}$ and $-1.05 \cdot 10^{-7}$. For the positive theorem, a two-mode argument proves global positivity of the log-curvature quantities q, F_2 and of the fully confluent invariant $\mathcal{C}_4$. The finite verification uses only rational polynomial algebra, Taylor inequalities, and geometric tails. The order-five origin, threshold, and finite-witness certificates likewise use exact rational arithmetic. A quotient-Wronskian argument reduces arbitrary fourth-order minors to $\partial_\xi \Psi < 0$. In the increasing curvature coordinate, its numerator is

$$\delta\Lambda \int_p^w \Delta(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2} \, \mathrm{d}t.$$

Here Δ is the difference of the CDFs of two explicitly normalized, curvature-tilted triangular densities. Its strict positivity follows from a one-crossing density ratio. The transport reductions and analytic sign estimates are given explicitly, with a table of the finite polynomial inequalities used for the three global signs. We then construct an explicit positive even Schwartz kernel which is itself strictly PF_4 but whose entire Fourier transform has nonreal zeros; the latter conclusion follows from a short exact discriminant inequality. This delimits the finite-order reach of Schoenberg's infinite-order transform correspondence.

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1 Introduction

A function $f : \mathbb{R} \to \mathbb{R}$ is PF_r when every translation-kernel minor

$$\det[f(x_i - y_j)]_{i,j=1}^k$$

is nonnegative for $1 \le k \le r$ and increasing node tuples. We call the property strict when the determinant is positive whenever both tuples are strictly increasing. This paper determines the exact finite Pólya-frequency order of the Riemann kernel; see [3] for modern background and Schoenberg's characterization. Related finite-order and Riemann-kernel work is placed in Section 2.

Theorem 1.1 (Strict global PF_4). *For every $1 \le k \le 4$ and strictly increasing real tuples $x_1 < \cdots < x_k$ and $y_1 < \cdots < y_k$,*

$$\det[\Phi(x_i - y_j)]_{i,j=1}^k > 0.$$

A parity factorization of the fully confluent derivative matrix at the origin gives a negative order-five limit. Divided differences then transfer that sign to equally spaced translation minors at sufficiently small positive spacing; see Theorem 11.2. Hence Theorem 1.1 has the following exact consequence.

Corollary 1.2. *The exact global Pólya-frequency order of* Φ *is four.*

The exact-order classification and the Fourier question are distinct. The following example marks a precise finite-order boundary for Schoenberg's infinite-order transform correspondence, even within the positive even Schwartz class.

Theorem 1.3 (Continuous PF_4 Fourier separator)**.** *There is an explicit positive even Schwartz function* f_* *which is strict* PF_4*, while its entire Fourier transform has nonreal zeros. Consequently continuous* PF_3 *and continuous* PF_4 *membership are each insufficient to force Fourier real-rootedness.*

The proof has two one-variable inputs and one exact transport mechanism. First, global inequalities $q > 0$ and $F_2 > 0$ imply strict PF_3 and positivity of a three-point functional Λ. Second, the fully confluent order-four invariant $\mathcal{C}_4$ is globally positive. Exact quotient identities reduce strict PF_4 to the monotonicity of a scalar three-point function Ψ. An increasing curvature coordinate then turns the derivative of Ψ into a positive crossing-kernel integral of $\mathcal{C}_4$.

The finite part of the first input consists of exact rational Bernstein coefficients on fixed algebraic domains, followed by analytic bounds for all remaining theta modes; Appendix B lists the transformed target polynomials, domains, degrees, minimum coefficients, derivative envelopes, and the surviving relative margins. The transport algebra is proved in the text, and the order-five calculation uses rational enclosures and elementary series bounds. The same recurrence gives a global order-five refinement: its confluent coefficient has one positive zero, enclosed in a rational interval of width 10^{-13}, and an explicit rational spacing gives a comfortably negative finite determinant. Neither refinement is needed for the strict-PF_4 theorem, but both make the first failure quantitatively reproducible. Theorem 1.3, proved in Section 12, supplies an explicit order-four counterexample rather than a finite-order sufficiency threshold.

2 Relation to earlier work

Schoenberg introduced Pólya-frequency functions and connected translation minors with bilateral Laplace transforms and variation-diminishing convolution operators [1, 3]. Karlin's monograph remains the standard systematic account of total positivity and its composition principles [2]; a modern review of PF functions and their preservers is given by Belton, Guillot, Khare, and Putinar [3].

Finite-order translation kernels have a substantially wider range than the infinite-order class. Khare's characterization of measurable TN_p functions for $p \geq 3$ provides a general finite-order setting [4]. The present proof instead derives a criterion specific to order four, using successive quotient derivatives and a curvature transport identity.

The theta normalization used here is the classical one in de Bruijn's study of the Riemann Fourier kernel [5]. For the same kernel, Dimitrov and Xu relate Wronskians of Fourier and Laplace transforms to correlation kernels and real-zero criteria [6], while Csordas and Varga obtain strict concavity and moment inequalities connected with the Riemann Hypothesis [7]. Grochenig explains the distinct Schoenberg correspondence relevant to the Riemann Hypothesis: the PF function there is recovered from the reciprocal of a Laguerre–Pólya entire function, rather than identified with the original Riemann kernel [8].

Michałowski first published an explicit negative 5×5 Toeplitz minor for this kernel, supported by an interval computation, together with single-configuration positivity checks at orders at most four [9]. Global PF_4 and a symbolic proof of the order-five leading sign are posed there as Open Problems 1 and 2. Theorem 1.1 resolves the first, and Lemma 11.1 resolves the second in strengthened form. Indeed, for the Toeplitz determinant

$$D_r(u_0, h) = \det[\Phi(u_0 + (i-j)h)]_{i,j=0}^{r-1},$$

Lemma 6.2 gives its leading coefficient as

$$C_r(u_0) = \frac{(-1)^{r(r-1)/2} V(0, \ldots, r-1)^2}{\prod_{k=0}^{r-1} (k!)^2} \det[\Phi^{(i+j)}(u_0)]_{i,j=0}^{r-1}.$$

For $r = 5$ the sign factor is positive, only derivatives through order eight occur, and the limit as $u_0 \to 0^+$ is a positive multiple of the parity-factorized determinant $\mathsf{H}_5 = CB < 0$ computed exactly in Lemma 11.1. Theorem 11.3 determines the complete sign change: $C_5(u_0) < 0$ precisely for $0 \le u_0 < u_*$, where $0.0622795266356 < u_* < 0.0622795266357$. The normalization in [9] uses $K_M(u) = \Phi(2u)/2$, so its corresponding center is $u_*/2 \in (0.0311397633178, 0.03113976331785)$.

Theorem 1.3 further shows where the infinite-order transform conclusion ceases to follow from finite-order positivity: even strict PF_3 or PF_4 does not control all Fourier zeros.

3 The kernel and smooth even continuation

Put

$$\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}, \qquad \Theta(t) = \vartheta(e^{2t}), \qquad H(t) = e^{t/2}\Theta(t).$$

The theta transformation $\vartheta(x) = x^{-1/2}\vartheta(1/x)$ gives

$$\Theta(-t) = e^t \Theta(t), \qquad H(-t) = H(t).$$

The series and all its derivatives converge locally uniformly, so H is real analytic and even. Define

$$\Phi(t) = \frac{1}{2}\left(H''(t) - \frac{1}{4}H(t)\right).$$

This is twice the common de Bruijn kernel normalization; multiplication by a positive constant changes none of the PF signs or strictness statements. Termwise differentiation gives, for every real t,

$$\Phi(t) = \sum_{n \ge 1} \left(4\pi^2 n^4 e^{9t/2} - 6\pi n^2 e^{5t/2}\right) e^{-\pi n^2 e^{2t}}. \tag{3.1}$$

Thus the positive-side formula often written with $|t|$ is the restriction of a real-analytic even function. In particular, smoothness at the origin is not inferred from numerical coverage.

For $t \ge 0$, the coefficient of the n-th exponential can be factored as

$$2\pi n^2 e^{5t/2}\left(2\pi n^2 e^{2t} - 3\right) > 0,$$

because $2\pi > 3$. Hence every term in (3.1) is positive and $\Phi(t) > 0$ on $[0, \infty)$. Evenness gives $\Phi(t) > 0$ on all of $\mathbb{R}$, so the following logarithm is globally defined.

Write

$$\ell = \log\Phi, \qquad s = \ell', \qquad q = -s' = -\ell''.$$

For $a < b$, define

$$A(a,b) = s(a) - s(b), \qquad M(a,b) = \frac{q(b) - q(a)}{A(a,b)}. \tag{3.2}$$

When $q > 0$, s is strictly decreasing and $A(a,b) > 0$.

4 Certified one-variable inputs

Define

$$F_1 = qq'' - (q')^2, \qquad F_2 = q^3 - F_1.$$

For the logarithmic cumulants $c_j = \ell^{(j)}$, set

$$\begin{aligned} &m_0 = 1, \quad m_1 = 0, \quad m_2 = c_2, \quad m_3 = c_3, \\ &m_4 = c_4 + 3c_2^2, \quad m_5 = c_5 + 10c_2c_3, \\ &m_6 = c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3, \end{aligned}$$

and

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^4. \tag{4.1}$$

Lemma 4.1 (Certified input)**.** *For the kernel* Φ*,*

$$q(t) > 0, \qquad F_2(t) > 0, \qquad \mathcal{C}_4(t) > 0 \quad (t \in \mathbb{R}).$$

The proof combines exact rational polynomial inequalities with analytic theta-tail bounds. For $u \geq 0$, put

$$x = \pi e^{2u}, \qquad \eta = e^{-3x}, \qquad R_0(z) = 2z - 3,$$

and recursively

$$R_{j+1}(z) = \left(\frac{5}{2} - 2z\right) R_j(z) + 2zR_j'(z).$$

After division by the positive first theta mode, the first two modes give the normalized jet entries

$$a_j(x,\eta) = \frac{R_j(x) + 4\eta R_j(4x)}{R_0(x)}.$$

Exact rational Bernstein coefficients on three fixed algebraic domains prove

$$q_{1,2} > 10, \qquad (F_2)_{1,2} > 1000, \qquad (\mathcal{C}_4)_{1,2} > 50{,}000{,}000.$$

Here the half-line certificates are applied directly to the cleared numerators of the shifted targets $q_{1,2} - 10$, $(F_2)_{1,2} - 1000$, and $(\mathcal{C}_4)_{1,2} - 50{,}000{,}000$. The complete contribution of modes $n \geq 3$ is bounded by one monotone polynomial-geometric estimate. Its changes in the same cleared sign numerators are smaller than the three displayed margins, both on $\pi \leq x \leq 5$ and on $x \geq 5$. Evenness completes the real line. All inequalities, including the derivative envelopes used on the tail, are derived in Appendix B. The exact replay uses no floating point, no interval subdivision, and no FLINT or Arb arithmetic.

5 Order three and the positive slope functional

For $\xi < m < r$, define

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \tag{5.1}$$

We give the full estimate used to prove its positivity.

Set $f_0 = q'/q$. Because $s' = -q$,

$$M(a, b) = \frac{\int_a^b q'(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t} = \frac{\int_a^b f_0(t)q(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t};$$

thus $M(a, b)$ is a q-weighted mean of f_0. Choose a point $u \in [\xi, m]$ where f_0 is minimal on the left interval and a point $v \in [m, r]$ where it is maximal on the right. The weighted-mean bounds and the ordering of the extremizers are

$$M(\xi, m) \geq f_0(u), \qquad M(m, r) \leq f_0(v), \qquad u \leq m \leq v.$$

Consequently

$$M(m, r) - M(\xi, m) \leq f_0(v) - f_0(u) \leq \int_\xi^r \max(f_0'(t), 0)\,\mathrm{d}t.$$

Since

$$f_0' = \frac{qq'' - (q')^2}{q^2} = \frac{F_1}{q^2}$$

and $A(\xi, r) = \int_\xi^r q(t)\,\mathrm{d}t$, equation (5.1) yields

$$\Lambda(\xi; m, r) \geq \int_\xi^r \left(q - \frac{\max(F_1, 0)}{q^2} \right) \mathrm{d}t \tag{5.2}$$

$$= \int_\xi^r \frac{q^3 - \max(F_1, 0)}{q^2}\,\mathrm{d}t = \int_\xi^r \frac{\min(q^3, F_2)}{q^2}\,\mathrm{d}t > 0. \tag{5.3}$$

The second equality uses $q^3 - \max(F_1, 0) = \min(q^3, q^3 - F_1) = \min(q^3, F_2)$; the last inequality follows from Lemma 4.1 and $q > 0$.

The same quantity is the logarithmic slope derivative in the normalized order-three determinant. Hence (5.3), together with strict log-concavity, proves strict global PF_3. We rederive the needed Wronskian form in the next section, where the quotient integral transfers these confluent signs to every ordered collocation minor.

6 Quotient-Wronskian reduction

Fix $y_1 < y_2 < y_3 < y_4$, put $u_j(t) = \Phi(t - y_j)$, and write $p_j = t - y_j$, so $p_4 < p_3 < p_2 < p_1$. Define

$$v_j = (u_j/u_1)', \qquad w_j = (v_j/v_2)'.$$

Strict log-concavity gives

$$v_j = \frac{u_j}{u_1} A(p_j, p_1) > 0 \qquad (j \geq 2).$$

The case $k = 2$ of Lemma 6.1 below therefore proves strict order two directly. Successive column division and differentiation give

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \tag{6.1}$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{w_4}{w_3}\right). \tag{6.2}$$

Let $\mathcal{T}$ denote simultaneous translation of all point arguments. From (3.2),

$$\mathcal{T} A(a, b) = q(b) - q(a), \qquad \mathcal{T} \log A(a, b) = M(a, b). \tag{6.3}$$

The sign convention in (6.3) will be used below. Since

$$\frac{v_j}{v_2} = \frac{u_j}{u_2} \frac{A(p_j, p_1)}{A(p_2, p_1)},$$

direct differentiation gives

$$g_j := \mathcal{T} \log(v_j/v_2) = A(p_j, p_2) + M(p_j, p_1) - M(p_2, p_1). \tag{6.4}$$

The exact weighted-mean split

$$A(p_j, p_1)M(p_j, p_1) = A(p_j, p_2)M(p_j, p_2) + A(p_2, p_1)M(p_2, p_1)$$

turns (6.4) into

$$g_j = \frac{A(p_j, p_2)}{A(p_j, p_1)} \Lambda(p_j; p_2, p_1) > 0. \tag{6.5}$$

Thus $w_j = (v_j/v_2)g_j > 0$, and (6.1) has the strict order-three sign.

Define

$$\Psi(\xi; m, r) = \Lambda(\xi; m, r) + \mathcal{T} \log \Lambda(\xi; m, r).$$

Because $w_j = (v_j/v_2)g_j$,

$$\mathcal{T} \log(w_4/w_3) = (g_4 + \mathcal{T} \log g_4) - (g_3 + \mathcal{T} \log g_3). \tag{6.6}$$

Equation (6.5) and (6.3) give

$$\mathcal{T} \log g_j = M(p_j, p_2) - M(p_j, p_1) + \mathcal{T} \log \Lambda_j,$$

where $\Lambda_j = \Lambda(p_j; p_2, p_1)$. A second use of the same mean split gives

$$g_j + M(p_j, p_2) - M(p_j, p_1) = \Lambda_j - A(p_2, p_1).$$

The common last term cancels in (6.6), yielding the exact identity

$$\mathcal{T} \log(w_4/w_3) = \Psi(p_4; p_2, p_1) - \Psi(p_3; p_2, p_1). \tag{6.7}$$

Lemma 6.1 (Iterated quotient integral)**.** *Let $f_1, \ldots, f_k$ be continuously differentiable on an interval, let $f_1 > 0$, and put $G_j = f_j/f_1$. For $t_1 < \cdots < t_k$,*

$$\det[f_j(t_i)]_{i,j=1}^k = \prod_{i=1}^k f_1(t_i) \int_{t_1}^{t_2} \cdots \int_{t_{k-1}}^{t_k} \det[G'_{j+1}(\tau_i)]_{i,j=1}^{k-1} \,\mathrm{d}\tau_1 \cdots \mathrm{d}\tau_{k-1}. \tag{6.8}$$

The identity may be applied repeatedly whenever the first member of the new derivative family is positive.

Proof. Factor $f_1(t_i)$ from row i. The first column is then one. Replace successive rows, from bottom to top, by their difference with the preceding row. Each remaining entry is $G_j(t_{i+1}) - G_j(t_i) = \int_{t_i}^{t_{i+1}} G_j'(\tau)\,\mathrm{d}\tau$. Determinant multilinearity gives (6.8). Since $\tau_i \in [t_i, t_{i+1}]$, the integration nodes remain ordered, so the same argument applies to the derivative determinant. □

Lemma 6.2 (Confluent divided-difference limit)**.** *Let $f \in C^{2k-2}$, let $V(a) = \prod_{0\le i<j<k}(a_j - a_i)$, and let a, b be fixed strictly increasing k-tuples. Then*

$$\lim_{h\downarrow 0} \frac{\det[f(z + h(a_i - b_j))]_{i,j=0}^{k-1}}{h^{k(k-1)}V(a)V(b)} = \frac{(-1)^{k(k-1)/2}}{\prod_{r=0}^{k-1}(r!)^2} \det[f^{(i+j)}(z)]_{i,j=0}^{k-1}. \tag{6.9}$$

If only the row nodes coalesce while $y_0 < \cdots < y_{k-1}$ remain fixed, then

$$\lim_{h\downarrow 0} \frac{\det[f(z + ha_i - y_j)]_{i,j=0}^{k-1}}{h^{k(k-1)/2}V(a)} = \frac{\det[f^{(i)}(z - y_j)]_{i,j=0}^{k-1}}{\prod_{r=0}^{k-1} r!}. \tag{6.10}$$

Proof. Apply successive Newton divided-difference row operations, whose divisors are the positive numbers $h(a_j - a_i)$, and let $h \downarrow 0$. This gives (6.10). Applying the same operation to the columns gives derivatives with respect to $-y$, hence the factor $(-1)^{0+\cdots+(k-1)}$, and proves (6.9). This records the factorial normalization and does not assume a nonzero leading coefficient. □

Proposition 6.3 (Three-point equivalence)**.** *Under the strict lower-order positivity already proved, global* PF_4 *is equivalent to*

$$\partial_\xi \Psi(\xi; m, r) \le 0 \qquad (\xi < m < r).$$

Strict inequality implies strict global PF_4.

Proof. If $\partial_\xi \Psi \le 0$, then $p_4 < p_3$ makes the right side of (6.7) nonnegative. All prefactors in (6.2) are positive. The Wronskian is therefore nonnegative, and is positive in the strict case.

For clarity, applying Lemma 6.1 three times to the order-four minor gives, with $I_i = [t_i, t_{i+1}]$,

$$\begin{aligned}\det[u_j(t_i)]_{i,j=1}^4 = \prod_{i=1}^4 u_1(t_i) \int_{I_1\times I_2\times I_3} \prod_{i=1}^3 v_2(\tau_i)\\ \times \int_{\tau_1}^{\tau_2}\int_{\tau_2}^{\tau_3} w_3(\sigma_1)w_3(\sigma_2)\int_{\sigma_1}^{\sigma_2}\left(\frac{w_4}{w_3}\right)'(\rho)\,\mathrm{d}\rho\,\mathrm{d}\sigma_2\,\mathrm{d}\sigma_1\,\mathrm{d}\boldsymbol{\tau}.\end{aligned}$$

Every factor is positive in the strict case, and every integration interval has positive length. This proves positivity of every ordered collocation minor, not only of the confluent Wronskian.

Conversely, assume global PF_4. Fix arbitrary $\xi_1 < \xi_2 < m < r$, choose any t, and choose the columns so that $(p_4, p_3, p_2, p_1) = (\xi_1, \xi_2, m, r)$ at t; explicitly, $y_j = t - p_j$, which preserves $y_1 < \cdots < y_4$. Coalesce arbitrary strictly ordered row nodes at t. Equation (6.10) and global PF_4 make the resulting translate Wronskian nonnegative. Equations (6.2) and (6.7) therefore give

$$\Psi(\xi_1; m, r) \ge \Psi(\xi_2; m, r).$$

Because both the pair $\xi_1 < \xi_2$ and the pair $m < r$ were arbitrary, $\Psi(\cdot; m, r)$ is nonincreasing for every admissible m, r. Smoothness makes this equivalent to the stated differential inequality. □

7 Curvature coordinate and sign bridge

Since $q > 0$,

$$y(t) = -s(t), \qquad Q(y(t)) = q(t)$$

defines a global strictly increasing coordinate on its image, with $\mathrm{d}y/\mathrm{d}t = q(t) = Q(y) > 0$. Primes in this and the next three sections mean y-derivatives. Put

$$\rho = 1 - Q'' = \frac{F_2}{q^3} > 0, \qquad \kappa = 2 - Q'' = 1 + \rho > 1.$$

For $\xi < m < r$, write

$$p = y(\xi), \quad z = y(m), \quad w = y(r), \quad L = z - p, \quad R = w - z.$$

Because $A(a, b) = y(b) - y(a)$ and $M(a, b) = Q[y(a), y(b)]$,

$$\Lambda = (w - p) + Q[p, z] - Q[z, w]. \tag{7.1}$$

Two integrations of $\kappa = 2 - Q''$ give

$$\Lambda = \frac{1}{L}\int_p^z (y - p)\kappa(y)\,\mathrm{d}y + \frac{1}{R}\int_z^w (w - y)\kappa(y)\,\mathrm{d}y, \tag{7.2}$$

$$\delta := -\partial_p\Lambda = \frac{1}{L^2}\int_p^z (z - y)\kappa(y)\,\mathrm{d}y > 0. \tag{7.3}$$

Simultaneous translation becomes

$$\mathcal{T} = Q(p)\partial_p + Q(z)\partial_z + Q(w)\partial_w,$$

and $\partial_\xi = Q(p)\partial_p$. Thus $\Lambda_\xi = -Q(p)\delta$.

Define

$$\mathcal{N} = \delta\Lambda^2 + Q'(p)\delta\Lambda + \Lambda\mathcal{T}\delta - \delta\mathcal{T}\Lambda. \tag{7.4}$$

Differentiating $\Psi = \Lambda + \mathcal{T}\log\Lambda$, commuting simultaneous translation with ∂_ξ, and collecting over Λ^2 gives

$$\partial_\xi\Psi = -\frac{Q(p)}{\Lambda^2}\mathcal{N}. \tag{7.5}$$

Equivalently,

$$\frac{\mathcal{N}}{\delta\Lambda} = K := \Lambda + Q'(p) + \mathcal{T}\log\delta - \mathcal{T}\log\Lambda. \tag{7.6}$$

It remains to prove $K > 0$.

8 The confluent invariant in curvature form

Define

$$D = 3(\kappa - 1) - \{Q(\log\kappa)'\}'. \tag{8.1}$$

Lemma 8.1 (Confluent-to-curvature identity)**.** *The determinant* (4.1) *satisfies*

$$\mathcal{C}_4 = Q^6\kappa^2 D. \tag{8.2}$$

Consequently $D > 0$ *on* $\mathbb{R}$.

Proof. The change of variable gives $\mathrm{d}/\mathrm{d}t = Q\,\mathrm{d}/\mathrm{d}y$ and $c_2 = -Q$. Recursively applying this operator produces the five cumulants listed in Appendix A. Substitution into the Schur-complement form of (4.1) factors out Q^6. The remaining factor is

$$\kappa^2\left(3(\kappa-1) - \{Q(\log\kappa)'\}'\right),$$

as shown by the common expanded numerator in Appendix A. This proves (8.2) without a numerical assumption. Lemma 4.1, $Q > 0$, and $\kappa > 0$ then imply $D > 0$. $\square$

The determinant definition is primary: it is the doubly confluent fourth-order translation minor after division by the positive row and column Vandermonde factors and by Φ^4. The explicit thirteen-term expansion is included in Appendix A for independent checking of the certificate input.

9 Exact transport identity

Normalize the triangular weights in (7.2) and (7.3) as probability measures:

$$\mathrm{d}\mu(y) = \frac{(z-y)\kappa(y)}{L^2\delta}\mathbf{1}_{[p,z]}(y)\,\mathrm{d}y, \tag{9.1}$$

$$\mathrm{d}\nu(y) = \frac{(y-p)\kappa(y)}{L\Lambda}\mathbf{1}_{[p,z]}(y)\,\mathrm{d}y + \frac{(w-y)\kappa(y)}{R\Lambda}\mathbf{1}_{[z,w]}(y)\,\mathrm{d}y. \tag{9.2}$$

Set

$$A_0(y) = 3(y - Q'(y)) - Q(y)\frac{\kappa'(y)}{\kappa(y)}.$$

Then $A_0' = D$.

Lemma 9.1 (Transport cancellation)**.** *With K as in* (7.6),

$$K = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]. \tag{9.3}$$

Proof. Two elementary primitives carry the complete cancellation:

$$\kappa A_0 = \mathcal{H}', \qquad \mathcal{H} = 3y^2 - 3Q - 3yQ' + (Q')^2 + QQ'', \tag{9.4}$$

$$\mathcal{H} = \mathcal{J}', \qquad \mathcal{J} = y^3 - 3yQ + QQ'. \tag{9.5}$$

Both identities follow by one differentiation.

Let $I_L = \mathcal{J}(z) - \mathcal{J}(p)$ and $I_R = \mathcal{J}(w) - \mathcal{J}(z)$. Integrating the linear weights in (9.1)–(9.2) by parts once gives

$$\mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0] = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{L\mathcal{H}(p) - I_L}{L^2\delta}. \tag{9.6}$$

On the other hand, differentiation of the triangular integrals along $\mathcal{T}$ gives

$$\mathcal{T}\Lambda = U(z,w) - U(p,z), \quad \mathcal{T}\delta = \frac{U(p,z) - (Q(z)-Q(p))\delta - Q(p)\kappa(p)}{L},$$

where

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d)-Q(c))^2}{(d-c)^2}. \tag{9.7}$$

Substitution in (7.6) yields

$$K = \Lambda + Q'(p) - Q[p,z] + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)\kappa(p)}{L\delta}. \tag{9.8}$$

The endpoint identity equating (9.6) and (9.8), with every substitution exposed, is Appendix C. It is an identity in the endpoint positions and the values of Q, Q', Q''; no bound or polynomial model is used. This proves (9.3). $\square$

10 The positive crossing kernel

Let F_μ, F_ν be the CDFs of (9.1)–(9.2), and put

$$\Delta(y) = F_\mu(y) - F_\nu(y).$$

Lemma 10.1 (Strict stochastic ordering). *One has* $\Delta(y) > 0$ *for* $p < y < w$*, while* $\Delta(p) = \Delta(w) = 0$.

Proof. On $[z, w]$, $F_\mu = 1$, so $\Delta = 1 - F_\nu > 0$ before w. On (p, z), the ratio of the two positive densities is

$$R_0(y) = \frac{\mathrm{d}\mu/\,\mathrm{d}y}{\mathrm{d}\nu/\,\mathrm{d}y} = \frac{\Lambda(z-y)}{L\delta(y-p)}. \tag{10.1}$$

Writing $C = \Lambda/(L\delta) > 0$, direct differentiation gives

$$R_0'(y) = -\frac{C(z-p)}{(y-p)^2} < 0. \tag{10.2}$$

Thus R_0 decreases strictly from $+\infty$ to 0. Hence $\Delta' = \mu' - \nu'$ changes sign exactly once, from positive to negative. Since $\Delta(p) = 0$ and

$$\Delta(z) = 1 - F_\nu(z) = \nu((z, w]) > 0,$$

the function cannot return to zero on $(p, z]$. The right interval was already handled. □

Integration by parts for CDFs, using $A_0' = D$ and the common total mass, turns Lemma 9.1 into

$$K = \int_p^w \Delta(y) D(y)\,\mathrm{d}y. \tag{10.3}$$

Combining (7.6), (8.2), and (10.3) gives the central identity

$$\mathcal{N} = \delta\Lambda \int_p^w \Delta(y) \frac{\mathcal{C}_4(y)}{Q(y)^6 \kappa(y)^2}\,\mathrm{d}y > 0. \tag{10.4}$$

Every factor in the interior is strictly positive.

11 Completion and exact order

Equation (7.5) and (10.4) imply

$$\partial_\xi \Psi(\xi; m, r) < 0 \qquad (\xi < m < r).$$

Proposition 6.3 therefore proves Theorem 1.1.

We now prove directly that the classification stops at order four. Put

$$\mathsf{H}_k = \det[\Phi^{(i+j)}(0)]_{i,j=0}^{k-1}, \qquad s_j = \Phi^{(j)}(0).$$

Since Φ is even, simultaneous parity permutations of the rows and columns give

$$\mathsf{H}_4 = AB, \qquad \mathsf{H}_5 = CB,$$

where

$$A = s_0 s_4 - s_2^2, \qquad B = s_2 s_6 - s_4^2,$$

and

$$C = \det \begin{pmatrix} s_0 & s_2 & s_4 \\ s_2 & s_4 & s_6 \\ s_4 & s_6 & s_8 \end{pmatrix}.$$

Lemma 11.1. *For the Riemann kernel,*

$$445 < A < 446, \qquad 189223 < B < 189228, \qquad -674000000 < C < -614000000.$$

In particular, $\mathsf{H}_4 > 0$ *and* $\mathsf{H}_5 < 0$*.*

Proof. For $x = \pi n^2 e^{2u}$, the n-th summand of $\Phi(u)$ is $e^{u/2-x}T_0(x)$, where $T_0(x) = 2x(2x-3)$. If its j-th derivative is $e^{u/2-x}T_j(x)$, then

$$T_{j+1} = 2xT_j' + (1/2 - 2x)T_j.$$

Thus

$$s_j = \sum_{n\geq 1} e^{-\pi n^2} T_j(\pi n^2).$$

The first three modes are enclosed by rational alternating-series bounds. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ supplies rational bounds for π, and argument division followed by the degree-29 and degree-30 alternating Taylor sums supplies rational bounds for each exponential.

For the omitted modes, let S_j be the sum of the absolute coefficients of T_j. Since $\deg T_j = j+2$, $3 < \pi < 22/7$, and $e^3 > \sum_{r=0}^{8} 3^r/r! > 20$,

$$\sum_{n\geq 4} |T_j(\pi n^2)| e^{-\pi n^2} \leq \frac{S_j (22/7)^{j+2} 4^{2j+4} 20^{-16}}{1 - (5/4)^{20} 20^{-9}} \qquad (j = 0, 2, 4, 6, 8).$$

All endpoints lie on $10^{-36}\mathbb{Z}$, and every arithmetic operation is rounded outward. Substitution into the displayed formulas gives the entirely rational sign chain

$$0 < 445 < A < 446, \qquad 0 < 189223 < B < 189228,$$

$$-674000000 < C < -614000000 < 0.$$

Thus $AB > 0$ and $CB < 0$ without a floating-point determinant. The full grid endpoints and the standard-library verifier are included in the reproducibility archive. □

Theorem 11.2. *The Riemann kernel is not* PF_5*.*

Proof. Apply Lemma 6.2 with $k = 5$, $z = 0$, and $a_i = b_i = i$. The sign factor in (6.9) is one, both Vandermonde factors are positive, and Lemma 11.1 makes the limit strictly negative. Hence $\det[\Phi((i-j)h)]_{i,j=0}^{4} < 0$ for every sufficiently small positive h. These are translation minors at strictly increasing nodes. □

For the equal-step family put

$$D_5(u,h) = \det[\Phi(u + (i-j)h)]_{i,j=0}^{4}, \qquad C_5(u) = \lim_{h\downarrow 0} \frac{D_5(u,h)}{h^{20}}.$$

Because $V(0,\ldots,4) = \prod_{j=1}^{4} j!$, the Vandermonde and factorial factors in Lemma 6.2 cancel, and therefore

$$C_5(u) = \det[\Phi^{(i+j)}(u)]_{i,j=0}^{4}. \tag{11.1}$$

Theorem 11.3 (Confluent order-five threshold). *There is a unique $u_* > 0$ such that $C_5(u_*) = 0$, and*

$$0.0622795266356 < u_* < 0.0622795266357.$$

Moreover $C_5(u) < 0$ for $0 \le u < u_*$, while $C_5(u) > 0$ for $u > u_*$.

Proof. Put $x = \pi e^{2u}$, $y = e^{-3x}$, and $z = e^{-8x}$. After removing the positive first-mode amplitude $2xe^{5u/2-x}$, the first three theta modes of the derivative jet are

$$b_j = P_j(x) + 4yP_j(4x) + 9zP_j(9x),$$

where

$$P_0(x) = 2x - 3, \qquad P_{j+1}(x) = \left(\frac{5}{2} - 2x\right) P_j(x) + 2xP_j'(x).$$

Thus the three-mode part of C_5, apart from a positive fifth power, is the closed determinant

$$G_3(x, y, z) = \det[b_{i+j}]_{i,j=0}^4. \tag{11.2}$$

Its exact expansion has 231 monomials. The path derivative is obtained without another expansion from

$$\frac{d}{du} = 2x(\partial_x - 3y\partial_y - 8z\partial_z).$$

For $j \le 9$, the omitted modes satisfy

$$|e_j(x)| \le \frac{S_j x^{j+1} 4^{2j+4} e^{-15x}}{1 - (5/4)^{22} e^{-9x}}, \tag{11.3}$$

where S_j is the sum of the absolute coefficients of P_j. Exact rational Bernstein bounds, preserving the common x-dependence of e^{-3x} and e^{-8x}, give $C_5 < 0$ from $u = 0$ through $u = 1/50$ and $C_5' > 0$ on the overlapping range from $u = 3/200$ through $x = 4$. Direct four-mode endpoint enclosures give opposite signs at $u = 0.0622795266356$ and $u = 0.0622795266357$. Four further Bernstein cells give $C_5 > 0$ for $4 \le x \le 40$, and the exact one-mode determinant plus (11.3) gives positivity for $x \ge 40$. These overlapping signs prove existence, uniqueness, and the stated classification. The domains and cell counts are listed in Appendix F. □

The same rational evaluator gives a finite, nonconfluent witness:

$$-1.08 \cdot 10^{-7} < \det[\Phi((i-j)211/2000)]_{i,j=0}^4 < -1.05 \cdot 10^{-7}. \tag{11.4}$$

This is a direct translation minor at distinct equally spaced nodes. The outermost center over all finite spacings is a separate two-variable extremal problem; no such global extremality is used here.

Theorem 1.1 and Theorem 11.2 prove Corollary 1.2.

Remark 11.4 (Riemann Hypothesis boundary). Schoenberg's transform theorem says that the bilateral Laplace transform of a PF_∞ function is the reciprocal of a Laguerre–Pólya entire function on its strip of convergence [1, 3]. The Fourier transform of Φ, up to the standard normalization, is the Riemann Ξ-function, which has known real zeros; hence Φ is unconditionally not PF_∞. The Schoenberg formulation related to the Riemann Hypothesis instead asks for the inverse transform of the reciprocal of Ξ to be a Pólya-frequency function [8]. Thus the finite-order classification proved here is a different statement. The next section further shows that strict PF_4 alone does not imply Fourier real-rootedness.

12 An explicit continuous PF_4 separator

We now prove Theorem 1.3. Put

$$c = \frac{1}{64}, \qquad B(w) = \sum_{k=-3}^{3} b_k w^k, \qquad (b_{-3}, \ldots, b_3) = (2, 10, 23, 30, 23, 10, 2),$$

and define

$$f_*(x) = e^{-cx^2} B(e^{2cx}) = \sum_{k=-3}^{3} b_k e^{ck^2} e^{-c(x-k)^2}. \tag{12.1}$$

The Gaussian-mixture expression proves positivity and Schwartz decay; palindromy of (b_k) proves evenness.

12.1 Strict PF_3

For fixed x, give $\mathsf{X} \in \{-3, \ldots, 3\}$ probability proportional to $b_k e^{2ckx}$, and write χ_j for its cumulants. Differentiation of the finite log-partition function gives

$$q = 2c - (2c)^2 \chi_2, \qquad q' = -(2c)^3 \chi_3, \qquad q'' = -(2c)^4 \chi_4. \tag{12.2}$$

If $\bar{x} = \mathbb{E}\mathsf{X}$, then $0 \le \mathbb{E}[(\mathsf{X}+3)(3-\mathsf{X})] = 9 - \chi_2 - \bar{x}^2$. Thus $0 \le \chi_2 \le 9$, and hence

$$q \ge 2c(1 - 18c) = 2c\frac{23}{32} > 0. \tag{12.3}$$

Moreover,

$$\begin{aligned} F_2 &= q^3 - \{qq'' - (q')^2\} \\ &= q^3 + q(2c)^4 \chi_4 + (2c)^6 \chi_3^2. \end{aligned} \tag{12.4}$$

The identity $\chi_4 = \mathbb{E}(\mathsf{X} - \mathbb{E}\mathsf{X})^4 - 3\chi_2^2$ gives $\chi_4 \ge -243$. Dropping the final nonnegative term in (12.4) gives

$$F_2 \ge q\{q^2 - 3888c^4\}.$$

At $c = 1/64$, the two coefficients after division by c^2 are

$$q^2/c^2 \ge \frac{529}{256}, \qquad 3888c^2 = \frac{243}{256}.$$

Thus $F_2 \ge qc^2(143/128) > 0$ globally. The weighted-mean criterion of Section 5 proves strict PF_3.

12.2 Exact order-four density

Let $\epsilon = 2c = 1/32$, $t = e^{\epsilon x}$, and

$$P(t) = t^3 B(t) = 2 + 10t + 23t^2 + 30t^3 + 23t^4 + 10t^5 + 2t^6.$$

For $E = t\frac{\mathrm{d}}{\mathrm{d}t}$, define

$$N_1 = tP', \qquad N_{j+1} = t(N_j' P - j N_j P'). \tag{12.5}$$

The quotient rule gives $E^j \log P = N_j / P^j$. Therefore, for $\ell_j = (\log f_*)^{(j)}$,

$$\ell_2 = \frac{-\epsilon P^2 + \epsilon^2 N_2}{P^2}, \qquad \ell_j = \frac{\epsilon^j N_j}{P^j} \quad (3 \le j \le 6). \tag{12.6}$$

Insert these jets into the central-moment determinant (4.1). Every monomial has derivative weight twelve, so every term may first be put over the common denominator P^{12}. The resulting numerator has the exact factor P^8. After cancellation,

$$\mathcal{C}_4[f_*](x) = \frac{\mathcal{G}(t)}{2^{54}P(t)^4}. \tag{12.7}$$

The exact expansion printed in Appendix E proves that $\mathcal{G}$ is palindromic of degree 24 and that all 25 integer coefficients are strictly positive. Hence $\mathcal{C}_4[f_*] > 0$ on $\mathbb{R}$.

The established inequalities $q > 0$ and $F_2 > 0$ also give

$$\kappa = 2 - Q'' = \frac{q^3 + F_2}{q^3} > 0.$$

The generic transport identity (10.4) now has strictly positive density and crossing weight for f_*. Thus $\partial_\xi \Psi < 0$, and Proposition 6.3 proves that f_* is strict PF_4.

12.3 Nonreal Fourier zeros

With $\widehat{f}(z) = \int_{\mathbb{R}} f(x)e^{-izx}\,\mathrm{d}x$, Gaussian translation gives

$$\widehat{f_*}(z) = \sqrt{\frac{\pi}{c}}e^{-z^2/(4c)} \sum_{k=-3}^{3} b_k e^{ck^2} e^{-ikz}. \tag{12.8}$$

Put $w = e^{-iz}$ and $u = w + w^{-1}$. The Laurent factor in (12.8) vanishes exactly when

$$R_c(u) = 2e^{9c}u^3 + 10e^{4c}u^2 + (23e^c - 6e^{9c})u + 30 - 20e^{4c} \tag{12.9}$$

vanishes. At $c = 1/64$, the exact calculation in Appendix E gives $\mathrm{disc}(R_c) < 0$. Thus the real cubic has a nonreal conjugate pair. For $|w| = 1$, however, $w + w^{-1}$ is real. The corresponding w-roots therefore lie off the unit circle and lift through $w = e^{-iz}$ to zeros with $\operatorname{Im} z \neq 0$. The Gaussian factor is zero-free. This completes the proof of Theorem 1.3.

13 Exact verification and reproduction

The repository supplies independent verifiers for the finite algebra and rational inequalities used above.

Object	Repository verifier	Arithmetic
Global $q, F_2, \mathcal{C}_4$	`scripts/verify_riemann_signs_exact.py`	rational polynomial
Quotient reduction	`scripts/verify_pf4_wronskian_reduction.py`	symbolic exact
Transport identity	`scripts/verify_pf4_transport_kernel.py`	symbolic exact
Origin order-five sign	`scripts/verify_pf5_origin.py`	rational enclosures
Three-mode order-five polynomial	`scripts/verify_pf5_threshold_symbolic.py`	symbolic exact
Order-five threshold and witness	`scripts/verify_pf5_threshold.py`	rational Bernstein
Continuous separator	`scripts/verify_continuous_pf4_separator.py`	rational polynomial

The global-sign verifier retains the first two theta modes exactly. It checks the 361 Bernstein and decay inequalities summarized in Appendix B, the 140 derivative-envelope coefficients, and the analytic geometric bounds for all modes $n \geq 3$. Its launcher selects SymPy's pure-Python rational domain and refuses a FLINT import. No floating-point comparison or interval subdivision decides a sign.

The origin order-five verifier is independent of that calculation. It uses integer arithmetic with outward rounding on $10^{-36}\mathbb{Z}$, Machin bounds for π, alternating Taylor bounds for three theta modes, and the geometric tail in Lemma 11.1. The threshold verifiers reconstruct the 231-monomial three-mode determinant, certify the unique crossing by correlated rational Bernstein cells, and directly enclose (11.4). The separator verifier reconstructs the central determinant, checks the exact P^8 cancellation, verifies all 25 coefficients in (12.7), and reproduces the rational discriminant inequality in Appendix E. The generated coefficient record has SHA-256

`d0f91259919c7f237ef65068041122656636ca990fe6ef41a146dd53eaa1dba9`.

The quotient and transport scripts check identities for generic smooth jets. Their numerical evaluations at selected points are regression tests only; all sign conclusions follow from the displayed identities and exact inequalities.

From a clean checkout with Python 3.13, SymPy 1.14.0, and mpmath 1.3.0, the single replay command is

`python scripts/replay_paper.py`.

It prints one line beginning `PASS` for each certificate boundary, followed by `status=paper evidence replay passed`. The SHA-256 of its complete standard output is

`5204f9fa12bf539edc245c257db285ab42e2041e4735f782a84852e5c7f0a592`.

Package versions and the complete artifact hashes are also recorded in `requirements-paper.txt` and `paper/RELEASE_MANIFEST.md`.

14 Declarations, availability, and computational provenance

Author contribution. Joshuah Rainstar conceived and directed the project, developed the proof architecture, verified the exact computations, and prepared the manuscript.

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Conflict of interest. The author declares no competing interests.

Code and data availability. The public repository is `https://github.com/falseywinchnet/zeta`. The exact analytic proof was promoted in commit `8a4db9e24b3a`. That immutable snapshot contains the submission source, coefficient records, environment lock, replay tool, and PDF; their SHA-256 values are listed in `paper/RELEASE_MANIFEST.md`. The complete repository release is assigned the DOI `https://doi.org/10.5281/zenodo.21404360`. The complete evidence replay is `python scripts/replay_paper.py` from the repository root.

All theorem-bearing sign decisions are exact. Numerical values produced by the repository scripts are diagnostic only; there is no separate experimental dataset.

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A Confluent and curvature algebra

The central-moment Hankel matrix is

$$\begin{pmatrix} 1 & 0 & m_2 & m_3 \\ 0 & m_2 & m_3 & m_4 \\ m_2 & m_3 & m_4 & m_5 \\ m_3 & m_4 & m_5 & m_6 \end{pmatrix}.$$

Taking the Schur complement of the upper-left entry reduces its determinant to

$$\det\begin{pmatrix} m_2 & m_3 & m_4 \\ m_3 & m_4 - m_2^2 & m_5 - m_2 m_3 \\ m_4 & m_5 - m_2 m_3 & m_6 - m_3^2 \end{pmatrix}.$$

Substitution of the moments from Section 4 and ordinary three-by-three expansion gives

$$\mathcal{C}_4 = 12c_2^6 + 24c_2^4c_4 - 24c_2^3c_3^2 + 2c_2^3c_6 - 12c_2^2c_3c_5 + 7c_2^2c_4^2 \tag{A.1}$$

$$+ 12c_2c_3^2c_4 + c_2c_4c_6 - c_2c_5^2 - 9c_3^4 - c_3^2c_6 + 2c_3c_4c_5 - c_4^3. \tag{A.2}$$

In the curvature coordinate, repeated application of $Q\,\mathrm{d}/\,\mathrm{d}y$ to $c_2 = -Q$ gives

$$\begin{aligned}
c_2 &= -Q,\\
c_3 &= -QQ',\\
c_4 &= -Q\{(Q')^2 + QQ''\},\\
c_5 &= -Q\{(Q')^3 + 4QQ'Q'' + Q^2Q'''\},\\
c_6 &= -Q\{(Q')^4 + 11Q(Q')^2Q'' + 4Q^2(Q'')^2 + 7Q^2Q'Q''' + Q^3Q''''\}.
\end{aligned}$$

Substitution in the preceding polynomial gives

$$\mathcal{C}_4 = -Q^6\mathcal{P},$$

where

$$\begin{aligned}
\mathcal{P} = {} & QQ''Q'''' - Q(Q''')^2 - 2QQ'''' + Q'Q''Q''' - 2Q'Q'''\\
& + 3(Q'')^3 - 15(Q'')^2 + 24Q'' - 12.
\end{aligned}$$

Now $\kappa = 2 - Q''$, and direct differentiation gives

$$\{Q(\log\kappa)'\}' = \frac{Q'\kappa' + Q\kappa''}{\kappa} - \frac{Q(\kappa')^2}{\kappa^2},$$

so, after multiplication by κ^2 and substitution of $\kappa' = -Q'''$, $\kappa'' = -Q''''$,

$$3(\kappa - 1) - \{Q(\log\kappa)'\}' = -\frac{\mathcal{P}}{(Q''-2)^2} = -\frac{\mathcal{P}}{\kappa^2}.$$

Therefore $\mathcal{C}_4 = Q^6\kappa^2 D$, proving (8.2) by an exposed common numerator.

B Exact proof of the certified input

This appendix proves Lemma 4.1. The positive-side theta series has j-th derivative

$$2\pi n^2 e^{5u/2 - n^2x} R_j(n^2x), \qquad x = \pi e^{2u},$$

where the polynomials R_j satisfy the recurrence in Section 4. Machin's identity and alternating series give $\pi > 157/50$; hence $R_0(x) = 2x - 3 > 0$.

B.1 Cleared sign polynomials

In addition to a_j, define the normalized remaining modes

$$e_j(x) = \sum_{n\ge 3} \frac{n^2 e^{-(n^2-1)x} R_j(n^2x)}{R_0(x)}, \qquad \widehat{a}_j = a_j + e_j.$$

For an indeterminate raw jet $(\widehat{a}_0, \ldots, \widehat{a}_6)$, put

$$N_q = \widehat{a}_1^2 - \widehat{a}_0\widehat{a}_2,$$

$$
\begin{aligned}
N_{F_2} = {} & -\widehat{a}_0^4\widehat{a}_2\widehat{a}_4 + \widehat{a}_0^4\widehat{a}_3^2 + \widehat{a}_0^3\widehat{a}_1^2\widehat{a}_4 - 2\widehat{a}_0^3\widehat{a}_1\widehat{a}_2\widehat{a}_3 + 2\widehat{a}_0^3\widehat{a}_2^3 \\
& - 3\widehat{a}_0^2\widehat{a}_1^2\widehat{a}_2^2 + 3\widehat{a}_0\widehat{a}_1^4\widehat{a}_2 - \widehat{a}_1^6,
\end{aligned}
$$

and

$$N_{\mathcal{C}_4} = \det[\widehat{a}_{i+j}]_{i,j=0}^3.$$

Direct logarithmic differentiation and the central-moment determinant give

$$q = \frac{N_q}{\widehat{a}_0^2}, \qquad F_2 = \frac{N_{F_2}}{\widehat{a}_0^6}, \qquad \mathcal{C}_4 = \frac{N_{\mathcal{C}_4}}{\widehat{a}_0^4}. \tag{B.1}$$

The thirteen-term cumulant expansion obtained from the last identity agrees with (A.1); this is an exact polynomial identity.

For orientation, retaining only the first mode gives

$$q_1 = \frac{4x(4x^2 - 12x + 15)}{(2x-3)^2},$$

$$(F_2)_1 = \frac{64x^3(64x^6 - 576x^5 + 2448x^4 - 6432x^3 + 10044x^2 - 8100x + 2295)}{(2x-3)^6},$$

$$(\mathcal{C}_4)_1 = \frac{49152x^6(16x^4 - 96x^3 + 360x^2 - 840x + 945)}{(2x-3)^4}.$$

After $x = 5 + z$, every numerator has positive coefficients. The second mode is retained exactly because it supplies the larger margin near the origin.

B.2 Exact two-mode margin

Rational Taylor bounds for the exponential imply

$$0 < \eta < \frac{1}{12000} \quad (x \geq \pi),$$

and

$$\frac{1}{23000} < \eta < \frac{1}{12000} \quad \left(\frac{157}{50} \leq x \leq \frac{10}{3}\right).$$

The close endpoints are entirely rational checks. Namely,

$$\sum_{k=0}^{16} \frac{(471/50)^k}{k!} > 12000, \qquad \sum_{k=0}^{21} \frac{10^k}{k!} > 22000,$$

and the geometric bound for the tail after degree 13 gives

$$e^{10} < \sum_{k=0}^{13} \frac{10^k}{k!} \frac{10^{14}/14!}{1 - 10/15} = \frac{3480060751}{154791} < 23000.$$

These prove, respectively, the upper η-cap, the cap $e^{-10} < 1/22000$, and the lower bound $e^{-10} > 1/23000$. Likewise $\sum_{k=0}^{21} 15^k/k! > 3{,}000{,}000$, which gives $\eta < 1/3{,}000{,}000$ for $x \geq 5$. We use the following elementary certificate lemma. If $p(s,t) = \sum p_{ij}s^it^j$ has bidegree (d,e), its Bernstein coefficients on the unit square are

$$\beta_{k\ell} = \sum_{i \leq k,\, j \leq \ell} p_{ij} \frac{\binom{k}{i}}{\binom{d}{i}} \frac{\binom{\ell}{j}}{\binom{e}{j}}.$$

The power-to-Bernstein identity proves that $\beta_{k\ell} > 0$ for every k, ℓ implies $p > 0$ on the square.

Apply this lemma after affine rational changes of variables. For q, one rectangle ends at $x = 10/3$, followed by a positive base polynomial minus exponentially decreasing corrections. For F_2, dependent bounds on the first rectangle are followed by a rectangle through $x = 5$ and a positive half-strip expansion. For $\mathcal{C}_4$, the first rectangle is followed by two negative odd-power corrections whose ratios to the positive base decrease. The derivative of each ratio has a numerator with positive coefficients after the indicated shift. The 361 resulting rational coefficient inequalities give

$$q_{1,2} > 10, \qquad (F_2)_{1,2} > 1000, \qquad (\mathcal{C}_4)_{1,2} > 50{,}000{,}000. \tag{B.2}$$

This is a finite algebraic proof on fixed domains.

Here is the precise connection between those coefficient checks and the quantitative margins. Evaluate the raw polynomials at the two-mode jet and set

$$\mathcal{Q} = N_q - 10a_0^2, \qquad \mathcal{F} = N_{F_2} - 1000a_0^6, \qquad \mathcal{C} = N_{\mathcal{C}_4} - 50{,}000{,}000a_0^4.$$

The rows labelled Q, F, C below certify the cleared numerators of $\mathcal{Q}, \mathcal{F}, \mathcal{C}$, respectively. Thus their positivity is exactly, rather than merely qualitatively, the three inequalities in (B.2). For a ratio row, write the shifted target as $B_0(x) + \sum_{j\geq 1} \eta^j B_j(x)$, where B_0 has positive shifted coefficients. Whenever B_j is adverse, put $D_j = -B_j$. Positivity of

$$3jD_jB_0 - D_j'B_0 + D_jB_0'$$

after the same shift proves that $e^{-3jx}D_j(x)/B_0(x)$ decreases. The displayed endpoint ratio is the sum of all such adverse ratios at the left endpoint; being less than one proves the target positive on the entire half-line. Coefficientwise nonnegative pieces only increase it.

For reference, the complete finite certificate is summarized below. Write $\mathcal{A} = (2x-3)+4(8x-3)\eta$, which is positive on every listed domain. Rows Q_0, F_0, F_1, C_0 are Bernstein certificates after the indicated affine change to the unit square; the remaining rows use coefficient positivity after a half-line shift. In the two "ratio" rows, the listed fraction is the sum of the adverse coefficient ratios and is strictly less than one.

ID	Variables and domain	Degree	Positive denominator	Exact lower datum
Q_0	$157/50 \leq x \leq 10/3,\ 0 \leq \eta \leq 1/12000$	$(4,2)$	$\mathcal{A}^2$	6613625187767/70312500000
Q_∞	$x = 10/3 + z,\ 0 \leq \eta \leq 1/12000$	$(4,2)$	$\mathcal{A}^2$	ratio 47333/505500
F_0	$157/50 \leq x \leq 10/3,\ 1/23000 \leq \eta \leq 1/12000$	$(12,6)$	$\mathcal{A}^6$	minimum $>$ 1610047
F_1	$10/3 \leq x \leq 5,\ 0 \leq \eta \leq 1/22000$	$(12,6)$	$\mathcal{A}^6$	29124973000/19683
F_∞	$x = 5 + z,\ \eta = v/3000000,\ 0 \leq v \leq 1$	$(12,6)$	$\mathcal{A}^6$	432/19073486328125
C_0	$157/50 \leq x \leq 10/3,\ 0 \leq \eta \leq 1/12000$	$(14,4)$	$\mathcal{A}^4$	minimum $>$ 9546672947
C_∞	$x = 10/3 + z,\ 0 \leq \eta \leq 1/12000$	$(14,4)$	$\mathcal{A}^4$	ratio 65989337396066791/77165720550000000

These seven checks comprise 361 exact rational inequalities. The derivative envelopes used below contribute 140 further Bernstein coefficients, of bidegrees $(j+1,1)$, $0 \leq j \leq 6$; their smallest coefficient is 44861/31250. Thus the compact domains and the differentiated tail majorants belong to the same exact certificate.

B.3 All later modes

For direct reconstruction of the derivative bounds, put $S_j(t) = 2^j(-1)^j R_j(t)$. The recurrence gives

$$
\begin{aligned}
S_0 &= 2t - 3,\\
S_1 &= 8t^2 - 30t + 15,\\
S_2 &= 32t^3 - 224t^2 + 330t - 75,\\
S_3 &= 128t^4 - 1440t^3 + 4232t^2 - 3270t + 375,\\
S_4 &= 512t^5 - 8448t^4 + 41408t^3 - 68096t^2 + 30930t - 1875,\\
S_5 &= 2048t^6 - 46592t^5 + 343040t^4 - 976320t^3 + 1008968t^2 - 285870t + 9375,\\
S_6 &= 8192t^7 - 245760t^6 + 2536960t^5 - 11109120t^4 + 20633312t^3 - 14260064t^2 + 2610330t - 46875.
\end{aligned}
$$

For $t \geq 27$, shifting $t = 27 + z$ gives positive coefficients in $(-1)^j R_j(t)$ and in $2^{j+1}t^{j+1} - (-1)^j R_j(t)$. Therefore

$$0 < (-1)^j R_j(t) \leq 2^{j+1}t^{j+1}, \qquad 0 \leq j \leq 6. \tag{B.3}$$

The derivative numerator of the normalized $n = 3$ term is another positive shifted polynomial, so its absolute value decreases for $x \geq 3$. For $n \geq 4$, successive majorants have ratio at most

$$\left(\frac{5}{4}\right)^{16} e^{-27} < 10^{-9}.$$

Rational Taylor lower bounds for e^{24}, e^{27}, e^{45} then yield, on $\pi \leq x \leq 5$,

$$(|e_0|, \ldots, |e_6|) < (7\cdot 10^{-9}, 4\cdot 10^{-7}, 2\cdot 10^{-4}, 7\cdot 10^{-4}, 3\cdot 10^{-2}, 1.1, 41). \tag{B.4}$$

Indeed, the normalized $n = 3$ term decreases on $x \geq 3$, and $e^{24} > 25{,}000{,}000{,}000$, so it is bounded by $3|R_j(27)|/25{,}000{,}000{,}000$. The sum of the $n \geq 4$ majorants is at most

$$\frac{2^{j+2}3^j 4^{2j+4}}{3 \cdot 10^{19}}.$$

The data entering these two expressions are

j	$\lvert R_j(27)\rvert$	chosen upper bound for $\lvert e_j\rvert$
0	51	$7 \cdot 10^{-9}$
1	5037/2	$4 \cdot 10^{-7}$
2	475395/4	$2 \cdot 10^{-4}$
3	42678141/8	$7 \cdot 10^{-4}$
4	3623251731/16	$3 \cdot 10^{-2}$
5	288709329165/32	11/10
6	21373315648611/64	41

For each row the later-mode expression is less than one thousandth of the $n = 3$ expression, and their sum is strictly below the chosen bound. This derives every constant in (B.4) from the printed recurrence and three rational exponential inequalities. The same Bernstein lemma gives

$$(|a_0|, \ldots, |a_6|) < (2, 5, 20, 200, 2000, 60000, 1000000).$$

For any one of the printed raw polynomials $N = \sum_\alpha c_\alpha X^\alpha$, the perturbation estimate used here is the explicit finite sum

$$|N(a+e) - N(a)| \le \sum_\alpha |c_\alpha| \left\{ \prod_{j=0}^{6} (A_j + E_j)^{\alpha_j} - \prod_{j=0}^{6} A_j^{\alpha_j} \right\}, \tag{B.5}$$

whenever $|a_j| \le A_j$ and $|e_j| \le E_j$. Substitution of the two displayed seven-tuples into (B.5) gives

$$|\Delta N_q| < 0.000405, \quad |\Delta N_{F_2}| < 37.959, \quad |\Delta N_{\mathcal{C}_4}| < 31{,}549{,}532. \tag{B.6}$$

For $x \ge 5$, (B.3) gives

$$|a_j| \le 2^{j+2} x^j, \qquad |e_j| \le 2^{j+2} 3^{2j+4} x^j e^{-8x}.$$

The raw weights of $N_q, N_{F_2}, N_{\mathcal{C}_4}$ are respectively $2, 6, 12$. Thus every perturbation majorant is a positive multiple of $x^w e^{-8x}$, or a faster-decaying term, with $w \le 12$; it decreases for $x \ge 5$. In (B.5) take

$$A_j = 2^{j+2} 5^j, \qquad E_j = \frac{2^{j+1} 3^{2j+4} 5^j}{10^{17}}.$$

The latter follows from the displayed tail bound and $\sum_{k=0}^{47} 40^k/k! > 2 \cdot 10^{17}$. Direct substitution gives

$$|\Delta N_q| < 6.49 \cdot 10^{-11}, \quad |\Delta N_{F_2}| < 0.03, \quad |\Delta N_{\mathcal{C}_4}| < 418{,}287. \tag{B.7}$$

To compare like quantities, note that

$$a_0 = 1 + 4\eta \frac{R_0(4x)}{R_0(x)} > 1,$$

and, by homogeneity,

$$N_{q,1,2} = q_{1,2} a_0^2, \qquad N_{F_2,1,2} = (F_2)_{1,2} a_0^6, \qquad N_{\mathcal{C}_4,1,2} = (\mathcal{C}_4)_{1,2} a_0^4.$$

Consequently (B.2) supplies the same respective lower margins 10, 1000, and 50,000,000 for the cleared numerators. Both perturbation bounds (B.6) and (B.7) are strictly smaller. Their scale relative to the corresponding cleared-numerator margin is recorded here:

target	surviving fraction on $\pi \le x \le 5$	surviving fraction on $x \ge 5$
q	> 0.9999595	> 0.99999999999351
F_2	> 0.962041	> 0.99997
$\mathcal{C}_4$	> 0.36900936	> 0.99163426

Thus even the least relative margin, the compact $\mathcal{C}_4$ bound, retains more than 36.9 percent of its certified two-mode numerator margin. Finally $\widehat{a}_0 = a_0 + e_0 > 0$, so (B.1) proves all three signs for $u \ge 0$. Evenness proves Lemma 4.1.

C Endpoint algebra for the transport cancellation

This appendix records the algebra suppressed by the phrase "simplifying the endpoint expression." Put

$$z = p + L, \qquad w = z + R, \qquad a = \frac{Q(z) - Q(p)}{L}, \qquad b = \frac{Q(w) - Q(z)}{R}.$$

Direct integration of $2 - Q''$ gives

$$\delta = \frac{L + Q'(p) - a}{L}, \qquad \Lambda = L + R + a - b. \tag{C.1}$$

Define

$$\mathcal{J}_y = y^3 - 3yQ(y) + Q(y)Q'(y),$$
$$\mathcal{H}_p = 3p^2 - 3Q(p) - 3pQ'(p) + Q'(p)^2 + Q(p)Q''(p),$$

and, for endpoint data,

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d) - Q(c))^2}{(d-c)^2}.$$

With $I_L = \mathcal{J}_z - \mathcal{J}_p$ and $I_R = \mathcal{J}_w - \mathcal{J}_z$, the identity needed in Lemma 9.1 is exactly

$$\begin{aligned} &\frac{-(\mathcal{J}_z - \mathcal{J}_p)/L + (\mathcal{J}_w - \mathcal{J}_z)/R}{\Lambda} + \frac{L\mathcal{H}_p - (\mathcal{J}_z - \mathcal{J}_p)}{L^2\delta} \\ &\quad = \Lambda + Q'(p) - a + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)(2 - Q''(p))}{L\delta}. \end{aligned}$$

To check it, substitute (C.1), replace z by $p + L$ and w by $p + L + R$, and multiply by the common denominator $L^2R\delta\Lambda$. The terms containing $Q'(w)$, then $Q'(z)$, then $Q''(p)$ cancel pairwise; the remaining endpoint values and position terms cancel after inserting $a = (Q(z) - Q(p))/L$ and $b = (Q(w) - Q(z))/R$. No differential equation for Q is used, so the identity holds for arbitrary smooth Q. Equivalently, regard the endpoint jets $Q(p), Q(z), Q(w), Q'(p), Q'(z), Q'(w), Q''(p)$ and p, L, R as independent indeterminates subject only to the displayed definitions of a, b, δ, and Λ. Clearing $L^2R\delta\Lambda$ first and then expanding gives the zero polynomial. Expanding the uncleared rational expression need not expose this cancellation.

D Certificate inventory

The finite and symbolic parts of the proof are recorded in the following plain-text source files. Each record states the exact claim, its inputs, and the corresponding verifier.

Claim	Proof record
Global $q, F_2, \mathcal{C}_4$, order three	`sources/riemann-signs-analytic-certificate.md`
Quotient and three-point equivalence	`sources/pf4-wronskian-reduction-certificate.md`
Transport identity and completion	`sources/pf4-transport-kernel-certificate.md`
Origin order-five obstruction	`sources/pf5-origin-certificate.md`
Confluent threshold and finite witness	`sources/pf5-threshold-certificate.md`
Continuous strict-PF_4 Fourier separator	`sources/continuous-pf4-separator-certificate.md`

The maintained source map `paper/PAPER_MAP.md` identifies the manuscript section in which each record is used. The single replay command in Section 13 checks the corresponding verifier outputs.

E Exact separator calculation

This appendix records the algebraic boundary behind (12.7). Starting from the cumulants $\ell_2, \dots, \ell_6$, the verifier reconstructs the central moments

$$m_0 = 1, \quad m_1 = 0, \quad m_2 = \ell_2, \quad m_3 = \ell_3,$$

$$m_4 = \ell_4 + 3\ell_2^2, \qquad m_5 = \ell_5 + 10\ell_3\ell_2,$$
$$m_6 = \ell_6 + 15\ell_4\ell_2 + 10\ell_3^2 + 15\ell_2^3,$$

and then forms $\det[m_{i+j}]_{i,j=0}^3$ directly. This independently regenerates the thirteen-term $\mathcal{C}_4$ polynomial; it does not import the expansion used for the Riemann kernel.

The recurrence (12.5) is then applied over $\mathbb{Q}[t]$. Substitution of (12.6) shows that each of the thirteen determinant monomials has common denominator P^{12}. After collecting, the numerator is divisible by P^8 in $\mathbb{Q}[t]$. Cancel this factor and clear the remaining powers of two, writing

$$\mathcal{C}_4[f_*](x) = \frac{\mathcal{G}(t)}{2^{54}P(t)^4}.$$

Exact rational arithmetic verifies

$$\deg \mathcal{G} = 24, \qquad [t^j]\mathcal{G} = [t^{24-j}]\mathcal{G} > 0 \quad (0 \le j \le 24),$$

with smallest integer coefficient

$$\min_{0 \le j \le 24} [t^j]\mathcal{G} = 3221225472.$$

Equivalently, the smallest coefficient before multiplication by 2^{54} is $3/2^{24}$. Write

$$\mathcal{G}(t) = \sum_{j=0}^{24} a_j t^j, \qquad a_j = a_{24-j}.$$

The 13 independent integer coefficients are:

j	a_j	j	a_j
0	3221225472	7	357704506872240
1	61435985920	8	701640591264080
2	575070935296	9	1172880923200640
3	3512452441920	10	1684288128586089
4	15702254784944	11	2088645133531440
5	54633824917920	12	2243243092164284
6	153607464071178		

This table is generated by `scripts/generate_separator_coefficients.py`. Its machine-readable companion `generated/separator-coefficients.json` has SHA-256 prefix `d0f91259919c7f23`; the complete hash is recorded in the release manifest. Coefficient positivity proves the sign on the full domain $t > 0$. Exact spot reconstructions at $t = 1/3, 1, 4$ independently compare the central determinant with $\mathcal{G}/(2^{54}P^4)$.

For the Fourier step, the identities

$$w + w^{-1} = u, \quad w^2 + w^{-2} = u^2 - 2, \quad w^3 + w^{-3} = u^3 - 3u$$

give (12.9) directly. Put

$$r = e^{1/64}, \qquad x = r^2 = e^{1/32}.$$

Exact expansion gives

$$\operatorname{disc}(R_c) = 4r^{10}G(x),$$

where

$$\begin{aligned} G(x) = {} & 432x^{13} - 4968x^9 + 900x^8 + 16200x^6 + 19044x^5 \\ & - 72600x^4 + 20000x^3 + 62100x^2 - 54334x + 13225. \end{aligned}$$

Writing $y = x - 1$ gives

$$\begin{aligned} G(1+y) = {} & -1 - 10y - 12y^2 + 680y^3 + 11532y^4 + 96660y^5 \\ & + 365400y^6 + 569664y^7 + 512172y^8 + 303912y^9 \\ & + 123552y^{10} + 33696y^{11} + 5616y^{12} + 432y^{13}. \end{aligned}$$

Moreover $0 < y < 1/30$. Indeed,

$$e < 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{18} = \frac{49}{18} < \frac{11}{4},$$

where $1/18 = (1/24)\sum_{j\geq 0} 4^{-j}$ bounds the tail beginning with $1/4!$. The binomial theorem gives

$$\frac{11}{4} < 1 + \frac{32}{30} + \frac{\binom{32}{2}}{30^2} + \frac{\binom{32}{3}}{30^3} = \frac{1891}{675} < \left(\frac{31}{30}\right)^{32}.$$

The sum of the positive terms at $y = 1/30$ is exactly

$$\sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} = \frac{1621193195302591}{3690562500000000} < 1.$$

The two omitted terms $-10y - 12y^2$ are negative. Consequently

$$G(1+y) < -1 + \sum_{j=3}^{13} \frac{[y^j]G(1+y)}{30^j} < 0,$$

and hence $\operatorname{disc}(R_c) < 0$ without numerical evaluation.

F Exact order-five threshold certificate

This appendix records the finite inequalities used in Theorem 11.3. All endpoints lie on $10^{-50}\mathbb{Z}$, and every product and rational scaling is rounded outward on that lattice. Machin's identity bounds π; positive Taylor sums with a geometric remainder bound e^{2u}; and argument division with alternating sums bounds the negative exponentials.

The recurrence in the theorem reconstructs $P_0, \ldots, P_{10}$ over $\mathbb{Q}[x]$. Independently expanding the determinant (11.2) gives a polynomial of degree 23 in x, degree at most five in each of y, z, and 231 nonzero monomials. On a cell $x = a + w$, $0 \leq w \leq d \leq 1/10$, the verifier keeps the correlation between the exponential variables by using

$$e^{-rx} = e^{-ra}\sum_{k=0}^{8} \frac{(-rw)^k}{k!} + E_r(w), \qquad |E_r(w)| \leq e^{-ra}\frac{(rd)^9}{9!}, \qquad r = 3, 8.$$

It substitutes these interval polynomials into G_3 and converts the result to the Bernstein basis in w. The later-mode boxes (11.3) are propagated through the 120 determinant monomials.

Object	Domain	Cells	Certified sign
C_5	$0 \le u \le 1/50$	2	negative
C_5'	$3/200 \le u,\ x \le 4$	22	positive
root endpoints	$u = 0.0622795266356,\ 0.0622795266357$	2	$(-,+)$
C_5	$4 \le x \le 40$	4	positive
one-mode tail	$x \ge 40$	1	positive
finite witness	$u = 0,\ h = 211/2000$	1	negative

At the two root endpoints the fourth theta mode is retained explicitly and only $n \ge 5$ is bounded. This gives the rational sign margins needed for the interval of width 10^{-13}. For $x \ge 40$, the one-mode determinant is

$$2 \cdot 150994944\, x^{10}(32x^5 - 240x^4 + 1200x^3 - 4200x^2 + 9450x - 10395).$$

After subtracting x^5, the polynomial has positive coefficients in $x - 40$. The remaining modes are smaller than this lower bound because $x^{10}e^{-3x}$ decreases there; the verifier checks the resulting exact rational ratio is below one.

The symbolic reconstruction is `scripts/verify_pf5_threshold_symbolic.py`. The independent lattice, Bernstein, tail, root, and finite-witness replay is `scripts/verify_pf5_threshold.py`.